



The Indian international School, DSO

Chemistry Department

PRACTICAL

Systematic Qualitative Analysis

Preliminary Analysis

S.NO.	EXPERIMENT	OBSERVATIONS	INFERENCE
1	Colour of the salt	Blue	Maybe Cu^{2+} is present
		Yellowish Brown	Maybe Fe^{3+} is present
		Green	May be Cu^{2+} or Ni^{2+}
		Light Pink, flesh colour, or earthy colour	Maybe Mn^{2+} is present
		White	All the colored salts may be absent
2	Flame test Mix a little of the salt with the con. HCl. Make a paste and take a small amount of the paste and introduce it into the flame.	Crimson red	Maybe Sr^{2+} is present
		Brick red	Maybe Ca^{2+} is present
		Apple green	Maybe Ba^{2+} is present
		Greenish blue	Maybe Cu^{2+} is present
3	Ash test Dissolve a little of the salt in con. HNO_3 . Then add Cobalt nitrate solution. Dip a filter paper in solution and then introduce it into the flame.	Green ash	May be Zn^{2+} present
		Blue ash	May be Al^{3+} present
		Pink ash	May be Mg^{2+} present

IDENTIFICATION OF ACID RADICAL(Anion)

S.NO.	EXPERIMENT	OBSERVATIONS	INFERENCE
1	Treat the salt with dil. HCl solution	Colorless and odorless gas with brisk effervescence evolves	CO_3^{2-} Present.
	A little of the salt is rubbed with dil. H_2SO_4	Vinegar smell	Acetate present
2	Treat with con. H_2SO_4 solution	Colorless gas with a pungent smell giving dense white fumes with glass rod dipped in NH_4OH .	Cl^- present
		Reddish brown vapor and solution turn brown	Br^- present
3	Treat a little of the salt with con. H_2SO_4 solution and add paper ball	Reddish brown vapor obtained on heating	NO_3^- present
4.	Add BaCl_2 to the salt solution	White ppt is formed which is insoluble in dil. HCl	SO_4^{2-} present

CONFIRMATIVE TEST FOR ACID RADICAL (Anion)

S.NO.	EXPERIMENT	OBSERVATIONS	INFERENCE
1	<u>Test for Carbonate</u> Add dil. HCl to the salt solution. Heat it.	Brisk effervescence of a colorless and odorless gas pass through limewater, it turns milky.	CO_3^{2-} Confirmed
2	Test for Acetate Dissolve a little salt in Water and add neutral FeCl_3 . Dilute the above solution and boil.	Reddish brown ppt	CH_3COO^- confirmed
2 (a)	<u>Test for Halogen</u> Salt + water. Add AgNO_3	White ppt soluble in NH_4OH	Cl^- confirmed
		Pale yellow ppt is partially soluble in NH_4OH	Br^- confirmed

2 (b)	<u>CHROMYL CHLORIDE TEST-</u> Mix a little salt with $K_2Cr_2O_7$ and conc. H_2SO_4 . Heat it. Pass the red vapors through NaOH. Add Acetic acid and lead acetate	Yellow ppt	Cl^- confirmed
2 (c)	Dissolve a little Salt in water. Add CCl_4 and Chlorine water and shake well.	Reddish brown layer	Br^- confirmed
5	Test for Nitrate To Salt solution, add freshly prepared $FeSO_4$ solution and con H_2SO_4 through the side of test tube.	Brown ring formed at the junction of two liquids	NO_3^- Confirmed
	Test for sulphate Salt + water. Add lead acetate.	White ppt is insoluble in dil. HCl but soluble in ammonium acetate solution	SO_4^{2-} confirmed

Fixing up the Group -Basic Radical (cation)

Group	EXPERIMENT	OBSERVATIONS	INFERENCE
Zero	Heat the given salt with NaOH	Smell of NH_3	Maybe NH_4^+ present

Group	EXPERIMENT	OBSERVATIONS	INFERENCE
1	To the original solution, add dil. HCl and Filter	ppt formed No ppt formed	Grp I (Pb^{2+}) present Grp I (Pb^{2+}) absent
2	Pass H_2S through the above solution.	ppt formed No ppt formed	Grp II (Cu^{2+}) present Grp II (Cu^{2+}) absent
3	To a little salt solution add Solid Ammonium chloride (NH_4Cl) and ammonium hydroxide (NH_4OH) solutions.	White gelatinous Ppt. Brown ppt No ppt formed	Grp-III (Al^{+3}) present Grp-III (Fe^{3+}) present Grp-III absent
4	Pass H_2S through the above solution.	Black ppt Flesh color ppt Dirty white ppt No ppt	Grp-IV (Ni^{2+}) present Grp IV (Mn^{2+}) present Grp IV (Zn^{2+}) present Grp IV Absent

5	To a little of the salt solution add solid NH₄Cl , NH₄OH and [(NH₄)₂CO₃] solutions.	A white ppt If no ppt formed.	Grp-V (Ba ²⁺ , Ca ²⁺ , Sr ²⁺) present Group V absent
6	Add NH₄Cl , NH₄OH , and Disodium hydrogen phosphate (Na₂HPO₄) . to the original solution. Scratch the inside of the test tube with the glass rod.	White crystalline ppt	Grp-VI (Mg ²⁺) present

CONFIRMATORY TEST FOR CATIONS

Group	EXPERIMENT	OBSERVATIONS	INFERENCE
Zero	1.Heat the given salt with NaOH and bring a glass rod dipped in Conc. HCl near the mouth of the test tube.	Evolution of dense white fumes.	NH₄⁺ confirmed
Zero	2.Add Nessler's reagent to the salt solution.	Brown ppt formed	NH₄⁺ confirmed
1	Analysis of group- I ppt Dissolve the ppt in hot water and divide it into 2 parts.		
	1. To 1 st part add potassium chromate solution.	Yellow ppt	Pb ²⁺ confirmed
	2.To 2 nd part add KI solution	Yellow ppt soluble on heating & reappears on cooling.	Pb ²⁺ confirmed
2	Analysis of group II ppt The group-II ppt is dissolved in 50% HNO ₃ & divided it into two parts.		
	1. To one part add NH ₄ OH	Pale blue ppt soluble in excess to form a deep blue solution.	Cu ²⁺ confirmed
	2. To 2 nd part, add acetic acid followed by potassium Ferro cyanide.	Chocolate brown ppt	Cu ²⁺ confirmed

3	Analysis of group – III ppt		
	1. Dissolve the ppt in dil. HCl and add a few drops of blue litmus and Add NH_4OH .	Blue ppt floating in a colorless solution	Al^{+3} confirmed
	2. Ash test: Dissolve a little of the salt in con. HNO_3 . Then add Cobalt nitrate solution. Dip a filter paper in solution and then introduce it into the flame.	Blue tinted ash	Al^{+3} confirmed
	(If the ppt is reddish-brown.) Dissolve ppt in dil. HCl & divide into two parts.		
	1) To one part add $\text{K}_4[\text{Fe}(\text{CN})_6]$ solution	Prussian blue colour	Fe^{3+} confirmed
	2) To the second part, add NH_4CNS	Blood red colour	Fe^{3+} confirmed
4	Analysis of group IV ppt		
	A. (If flesh colour ppt) Dissolve the ppt in dil. HCl and divide it into two parts.		
	1. To one part, add NaOH	White ppt that slowly turns brown and finally black in colour	Mn^{2+} confirmed
	2. To the second part, add PbO_2 and conc. HNO_3 Boil and cool. Observe the colour.	Purple colour	Mn^{2+} confirmed
	B. (If dirty white ppt) Dissolve the ppt in dil. HCl and divide it into 2 parts.		
	1. To one part, add NaOH	White ppt soluble in an excess solution of NaOH	Zn^{2+} confirmed
	2. To the second part add potassium ferrocyanide $\text{K}_4[\text{Fe}(\text{CN})_6]$ solution	Bluish white ppt	Zn^{2+} confirmed
	Ash test: Dissolve a little of the salt in con. HNO_3 . Then add Cobalt nitrate solution. Dip a filter paper in solution	Green Ash is formed	Zn^{2+} confirmed

	and then introduce it into the flame.		
5	Analysis of group-v ppt Dissolve the ppt in acetic acid and warm it. Divide it into 3 parts.		
	1.To one part , add potassium chromate	Yellow ppt	Ba ²⁺ confirmed
	Flame Test:	Apple green	Ba ²⁺ confirmed
	2.To the second part , add Ammonium sulphate solution	White ppt	Sr ²⁺ confirmed
	Flame Test:	Crimson red	Sr ²⁺ confirmed
	To the third part , add ammonium oxalate. Add a little Ammonium Hydroxide and scratch the side.	White ppt	Ca ²⁺ confirmed
	Flame test:	Brick red	Ca ²⁺ confirmed
6	Analysis of group-VI Add NaOH to the Original Solution, followed by a few drops of Magneson Reagent	Blue coloration or blue precipitate	Mg ²⁺ confirmed
	Ash test	Pink colored Ash	Mg ²⁺ confirmed

Reference:

[Comprehensive Practical Chemistry Xii Cbse | PDF](#)

[A Handbook of Practical Chemistry Class XII - Dr M L Sharma, Dr P N](#)

[Chaudhary, Dr P S Saud | Thuprai](#)